

# Syllabus

## Spring semester 2026

| Course Title | Calculus 2     |           | Course Code |                | MC3109   |  | Credit | 4                                                            | Classroom | 형설관3406/3407 |
|--------------|----------------|-----------|-------------|----------------|----------|--|--------|--------------------------------------------------------------|-----------|--------------|
|              |                |           | Type        |                | elective |  | Hours  | 5                                                            |           |              |
| Teacher      | Name           | Class no. | Office      | Office hours   | Contact  |  |        | E-mail                                                       |           |              |
|              | Jung Eun Kim   | EC2       | 3503        | Wed 2,3, Fri 2 | 606-2386 |  |        | <a href="mailto:Jkim.math@ksa.hs.kr">Jkim.math@ksa.hs.kr</a> |           |              |
|              | Jerome Tambour | EC1       | 3507        | Mon 6,7 Fri 3  | 606-2148 |  |        | <a href="mailto:jerome@ksa.hs.kr">jerome@ksa.hs.kr</a>       |           |              |

## 1. Course Descriptions

| Course Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>◆ To introduce students to understand integration, differential equations, series, derivatives and integrals of vector functions and to solve real world problem related to them.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>1. Further Applications of Integration</p> <ul style="list-style-type: none"> <li>i. Arc length</li> <li>ii. Area of a Surface of Revolution</li> <li>iii. Applications to Physics and Engineering</li> <li>iv. Probability</li> </ul> <p>2. Differential Equations</p> <ul style="list-style-type: none"> <li>i. Modelling with Differential Equations</li> <li>ii. Direction Fields and Euler's Method</li> <li>iii. Separable Equations</li> <li>iv. Models for Population Growth</li> <li>v. Linear Equations</li> </ul> <p>3. Parametric Equations and Polar Coordinates</p> <ul style="list-style-type: none"> <li>i. Curves Defined by Parametric Equations</li> <li>ii. Calculus with Parametric Curves</li> <li>iii. Polar Coordinates</li> <li>iv. Areas and Lengths in Polar Coordinates</li> <li>v. Conic Sections</li> <li>vi. Conic Sections in Polar Coordinates</li> </ul> | <p>4. Infinite Sequences and Series</p> <ul style="list-style-type: none"> <li>i. Sequences</li> <li>ii. Series</li> <li>iii. The Integral Test and Estimates of Sums</li> <li>iv. The Comparison Tests</li> <li>i. Alternating Series and Absolute Convergence</li> <li>v. The Ratio and Root Tests</li> <li>vi. Strategy for Testing Series</li> <li>vii. Power Series</li> <li>viii. Representations for Functions as Power Series</li> <li>ix. Taylor and Maclaurin Series</li> <li>x. Applications of Taylor Polynomials</li> </ul> <p>5. Vector Functions</p> <ul style="list-style-type: none"> <li>i. Vector Functions and Space Curves</li> <li>ii. Derivatives and Integrals of Vector Functions</li> <li>iii. Arc Length and Curvature</li> <li>iv. Motion in Space: Velocity and Acceleration</li> </ul> |

## 2. Core Target Competencies

| Subject Core Competencies                                                                                                                                                                                                                                     |                                |               |                         |                      |                         |                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------|-------------------------|----------------------|-------------------------|------------------------|
| Comprehension<br>Adaptability                                                                                                                                                                                                                                 | Problem-<br>Solving<br>Ability | Communication | Analytical<br>Reasoning | Creative<br>Thinking | Integrative<br>Thinking | Application<br>Ability |
| <b>Comprehension, Adaptability, Problem-Solving Ability (Assignments, Performance Assessment):</b><br>Understanding basic concepts of the subject and applying learned content appropriately in given situations.                                             |                                |               |                         |                      |                         |                        |
| <b>Communication, Analytical Reasoning, Creative Thinking (Assignments, Performance Assessment, Written Exams):</b><br>Evaluating the ability to analyze problems, creatively solve them by applying learned content, and logically articulate the solutions. |                                |               |                         |                      |                         |                        |
| <b>Integrative Thinking, Application Ability (Assignments, Written Exams):</b><br>Assessing the ability to solve problems in various fields such as physics, statistics, and geometry by utilizing calculus.                                                  |                                |               |                         |                      |                         |                        |

## 3. Textbooks and References

A. Textbook: Calculus, James Stewart, D. Clegg, S. Watson, 9th Edition, ISBN 978-0-357-11346-2

B. References: Thomas' CALCULUS, Finney, Weir and Giordano, ISBN 0134438981

## 4. Class Method

A: Lecture

B: Solving problems

C: Presentation

## 5. Evaluation

| Activities &<br>Participation | Quizzes | Mid-term Exam | Final Exam | Attendance<br>& Attitude | Total |
|-------------------------------|---------|---------------|------------|--------------------------|-------|
| 8%                            | 12%     | 35%           | 35%        | 10%                      | 100%  |
| 40                            | 60      | 175           | 175        | 50                       | 500   |

### 1) Activities

Activities include homework, in-class assignments and presentations.

### 2) Quizzes

There will be two in-class quizzes.

### 3) Exams

Mid-term : Chapter 8~10 Final : Chapter 11, 13 (It can be changed.)

### 3) Attendance

Absence: -3 points except for illness or official cases

(with confirmation from our school infirmary and contact in advance)

Lateness : -1 point

## 6. Criteria for Awarding Credits

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

● **Attendance for at least three-quarters of the total class hours is required to earn credit.**

● **Calculation of the basis score for awarding credit to absentees:** The score for an exam missed due to an official or medical absence is determined by taking the student's percentile score from their other exams in this subject. We find the score which corresponds to the same percentile for the missed exam. This score is then multiplied by a factor based on the reason for absence (100% for an official absence, 80% for a medical absence).

● **If a basis score cannot be calculated, the grading method:** If there is no basis score available, a make-up examination shall be conducted.

## 7. Weekly Plans

| Week (period)                | Topic and Contents                                                                                              | (Notes)                      |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Week 1</b><br>2.23.~3.1.  | Course introduction<br>8.1 Arc length<br>8.2 Area of a Surface of Revolution                                    |                              |
| Core Competencies            | ♦ Problem solving ability, application ability                                                                  |                              |
| <b>Week 2</b><br>3.2.~3.8.   | 8.3. Applications to Physics and Engineering<br>8.5. Probability (Random variables, Expectation)                | <b>3.2.(Mon)<br/>Holiday</b> |
| Core Competencies            | ♦ Problem solving ability, application ability, integrative thinking                                            |                              |
| <b>Week 3</b><br>3.9.~3.15.  | 8.5 Probability (Variance, Bernoulli distributions, Normal distributions)                                       |                              |
| Core Competencies            | ♦ Application ability, integrative thinking, analytical reasoning                                               |                              |
| <b>Week 4</b><br>3.16.~3.22. | 9.1 Modelling with Differential Equations<br>9.2 Direction Fields and Euler's Method<br>9.3 Separable Equations |                              |
| Core Competencies            | ♦ Comprehension adaptability, communication                                                                     |                              |
| <b>Week 5</b><br>3.23.~3.29. | 9.4 Models for Population Growth<br>9.5 Linear Equations<br>10.1 Curves Defined by Parametric Equations         | Quiz 1                       |
| Core Competencies            | ♦ Creative thinking, comprehension adaptability                                                                 |                              |
| <b>Week 6</b><br>3.30.~4.5.  | 10.2 Calculus with Parametric Curves<br>10.3 Polar Coordinates<br>10.4 Areas and Lengths in Polar Coordinates   |                              |
| Core Competencies            | ♦ Analytical reasoning, problem solving ability                                                                 |                              |
| <b>Week 7</b><br>4.6.~4.12.  | 10.5 Conic Sections<br>10.6 Conic Sections in Polar Coordinates<br>Review                                       |                              |
| Core Competencies            | ♦ Comprehension adaptability, communication                                                                     |                              |

|                               |                                                                                                                                        |                                             |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Week 8</b><br>4.13.~4.19.  | Midterm Exams <b>(4.14~4.17.)</b>                                                                                                      | <b>4.13.(Mon)</b><br><b>Regular Classes</b> |
| <b>Week 9</b><br>4.20.~4.26.  | 11.1 Sequences<br>11.2 Series                                                                                                          |                                             |
| <b>Week 10</b><br>4.27.~5.3.  | 11.3 The Integral Test and Estimates of Sums<br>11.4 The Comparison Tests                                                              |                                             |
| <b>Core Competencies</b>      | ◆ Comprehension adaptability, analytical reasoning                                                                                     |                                             |
| <b>Week 11</b><br>5.4.~5.10.  | 11.5 Alternating Series and Absolute Convergence<br>11.6 The Ratio and Root Tests                                                      |                                             |
| <b>Core Competencies</b>      | ◆ Application ability, problem solving ability                                                                                         |                                             |
| <b>Week 12</b><br>5.11.~5.17. | 11.7 Strategy for Testing Series<br>11.8 Power Series                                                                                  |                                             |
| <b>Core Competencies</b>      | ◆ Application ability, problem solving ability                                                                                         |                                             |
| <b>Week 13</b><br>5.18.~5.24. | 11.9 Representations for Functions as Power Series<br>11.10 Taylor and Maclaurin Series                                                | Quiz 2                                      |
| <b>Core Competencies</b>      | ◆ Creative thinking, integrative thinking                                                                                              |                                             |
| <b>Week 14</b><br>5.25.~5.31. | 11.11 Applications of Taylor Polynomials<br>13.1 Vector Functions and Space Curves                                                     | <b>5.25.(Mon)</b><br><b>Holiday</b>         |
| <b>Core Competencies</b>      | ◆ Creative thinking, integrative thinking                                                                                              |                                             |
| <b>Week 15</b><br>6.1.~6.7.   | 13.2 Derivatives and Integrals of Vector Functions<br>13.3 Arc Length and Curvature<br>13.4 Motion in Space: Velocity and Acceleration | <b>6.3.(Wed)</b><br><b>Election day</b>     |
| <b>Core Competencies</b>      | ◆ Application ability, analytical reasoning                                                                                            |                                             |
| <b>Week 16</b><br>6.8.~6.14.  | Review                                                                                                                                 | Make-up class week                          |
| <b>Week 17</b><br>6.15.~6.19. | Final Exams <b>(6.18~6.19.)</b>                                                                                                        |                                             |

※ The above plans may change according to the academic schedule.

## 8. Core Problem List

| Chapter   | Problem Number                                                                   |
|-----------|----------------------------------------------------------------------------------|
| 8.1       | 9, 13, 17, 39, 53                                                                |
| 8.2       | 10, 19, 20, 33, 37, 39, 46                                                       |
| P613      | Discovery project 1, 2, 3, 4, 5                                                  |
| 8.3       | (Example 1, 2) 26, 31, 44, 47, 48, 50                                            |
| 8.5       | 17, 18, 20                                                                       |
| P638~639  | 4, 9, 12, 14, 21, 23                                                             |
| P640~641  | 1, 2, 4, 5, 9, 10                                                                |
| 9.1       | 13, 15, 19, 21, 23, 29                                                           |
| 9.2       | 2, 11, 18, 22, 28                                                                |
| 9.3       | 7, 19, 24, 35, 41, 43, 57                                                        |
| 9.4       | 5, 7, 9, 11, 13, 19, 25                                                          |
| 9.5       | 8, 10, 11, 19, 23, 27, 28, 38                                                    |
| P694      | Concept Check 5, 6, 7, True-False Quiz 2, 7, 9                                   |
| P695~P696 | 4, 6, 10, 13, 18, 20                                                             |
| P697~698  | 3, 4, 9, 13                                                                      |
| 10.1      | 14, 19, 20, 21, 27, 28, 30, 34, 51, 52, 56, Lab. project 1                       |
| 10.2      | 9, 12, 20, 23, 35, 42, 49, 56, 72, 77, 79, 80, 81, 84                            |
| 10.3      | 3, 39, 43, 46, 47, 48,                                                           |
| 10.4      | 4, 8, 13, 21, 22, 27, 29, 31, 32, 35, 36, 39, 40, 42, 43, 44, 54, 67, 70, 75, 76 |
| 10.5      | 58, 59, 63, 65, 66                                                               |
| 10.6      | 17, 21, 22, 27, 30,                                                              |
| P760      | 1, 2, 4, 5, 6                                                                    |
| 11.1      | 74, 76, 85, 89, 90, 95, 97, 98                                                   |
| 11.2      | 61, 66, 70, 72, 77, 83, 84, 85, 86, 87, 88, 89, 90, 93, 95, 96                   |
| 11.3      | 32, 34, 46, 47                                                                   |
| 11.4      | 32, 33, 34, 37, 40, 45, 46, 47, 48, 49, 50, 51                                   |
| 11.5      | 29, 40, 46, 48, 49, 50                                                           |
| 11.6      | 17, 24, 25, 35, 40, 41, 45, 46                                                   |
| 11.7      | 2, 4, 6, ..., 48 (problems numbered even)                                        |
| 11.8      | 17, 19, 34, 35, 36, 37, 38, 39, 40, 41, 43, 45                                   |
| 11.9      | 1, 2, 14, 15, 38, 40, 41, 42, 46, 47, 50                                         |
| 11.10     | 2, 3, 4, 5, 11, 14, 16, 23, 30, 48, 61, 62, 79, 81, 83, 84, 96, 97, 98           |
| 11.11     | 5, 15, 19, 30, 39                                                                |
| P859~860  | 5, 15, 9, 30, 39                                                                 |
| P861      | 59, 60(a), 62                                                                    |
| P863~866  | 1, 2, 4, 5, 7, 8, 11, 12, 14, 15, 16, 20, 22, 23                                 |
| 13.1      | 1, 4, 9, 13, 15, 21~26. 35, 36, 51, 55, 58, 61(c)                                |
| 13.2      | 1, 3, 8, 18, 23, 26, 41, 43, 51, 55                                              |
| 13.3      | 4, 6, 14, 16, 18, 19, 26, 34, 46, 54, 64, 73                                     |
| 13.4      | 1, 9, 17, 21, 25, 29, 30, 33, 39, 45                                             |
| P965      | True-False 6, 8, 9, 13                                                           |
| P966~967  | 9, 11, 19, 23, 24                                                                |
| P968~970  | 1, 2, 3, 4, 5, 7, 9                                                              |